

Movie Streaming Analytics on Databricks

This project demonstrates a production-grade Databricks Lakehouse implementation that transforms raw streaming events into curated, governed datasets. By leveraging Delta pipelines, Unity Catalog, and AI/BI Genie spaces, the platform accelerates decision-making and establishes an agile foundation for AI-powered analytics in digital ecosystems.

01 BUSINESS CHALLENGE

EXECUTIVE BRIEF

- Data Fragmentation:** Massive user interaction logs (views, clicks, searches) fragmented across isolated systems without a unified layer.
- Operational Friction:** Lack of standardized data quality, governance, and slow execution of basic analytics queries by product teams.
- Insight Bottleneck:** Limited visibility into user engagement and content performance, leading to slower decision-making and delayed product optimization..

Why It Matters: Without unified analytics, platforms struggle to optimize content recommendations and retain users, directly impacting competitive personalization capabilities.

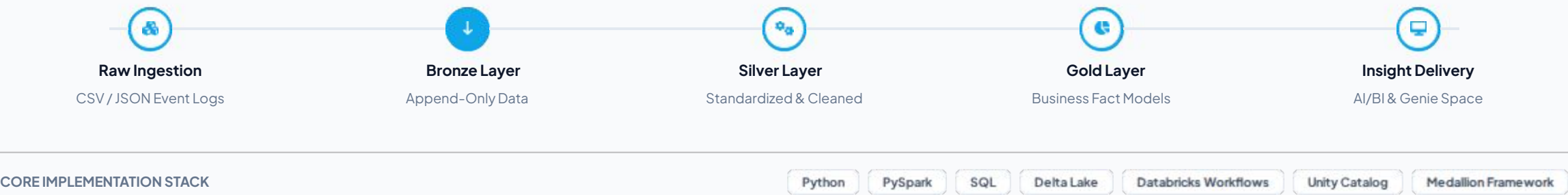
02 SOLUTION OVERVIEW

LAKEHOUSE PLATFORM

- Medallion Data Architecture:** Complete Bronze, Silver, Gold layers configured for high-volume ingest, transform, and analytics.
- Batch & Stream Ingestion:** Continuous JSON & CSV ingestion from object stores managed through Databricks Auto Loader.
- Democratic Consumption:** Unified catalog mappings enabling instant query resolution through Genie natural language spaces.

Core Impact: Shifts the organizational reporting dynamic from heavy engineering request backlogs to real-time self-service business intelligence.

03 SOLUTION ARCHITECTURE & DATA PIPELINE



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Project Type: Data Engineering | Lakehouse Architecture | Medallion Framework | Databricks Platform

04 VALUE REALIZATION MATRIX

STRATEGIC OUTCOME

OPERATIONAL METRIC AREA	TARGET STATE IMPACT
Data Reliability	Improved
Time to Insight	Reduced Latency
Analytics Scalability	Increased Velocity
Data Governance	Strengthened
Engineering Effort / Latency	Drastically Diminished

05 STRATEGIC VISION

TRANSFORMATION SCOPE

STRATEGIC SHIFT:



TARGET INDUSTRIES:

- Media & Entertainment
- OTT Platforms
- Subscription Services
- Gaming Analytics

06 FUTURE SCALE & PRODUCT INTEGRATION ROADMAP



Phase 1: Ingestion Base

Medallion Batch + Stream Setup



Phase 2: Near Real-Time

Monitoring & Incremental Pipelines



Phase 3: Prediction Layer

Advanced ML Feature Store



Phase 4: Agentic AI

Conversational BI Insights

INTEGRATION & ORCHESTRATION SYSTEMS

- Delta Live Tables
- Unity Catalog Governance
- AI/BI Genie Spaces
- Agentic BI Systems